

Circuits of Salt: Commodity Networks and the Politics of Trade in 20th Century Himalayas

Samagra Bharti

Human Sciences Research Center

International Institute of Information Technology, Hyderabad

Email: samagra.bharti@research.iiit.ac.in

Abstract

Salt has been one of the basic pillars of Himalayan and Tibetan life. The element has sustained mountain economies, shaped ritual practices, been the backbone of multi-commodity networks and even used as an exchange token by the drokpas and rupshu nomads. Our proposed paper tries to explore the politics of the salt economy in the Himalayas during the 20th Century. Studied mostly in its economic and anthropological aspects, salt in the Himalayas has been a tool to negotiate power, governance and territorial control. The study uses different historical archives, colonial gazetteers, administrative reports of regions across the Western Himalayas to develop a social-science literature around salt. Further, language-processing methods such as Named Entity Recognition along with relational-extraction models will be employed to attain a well-defined relational graph of salt along with various entities, commodities, communities, and administrative blocks. The political conceptualisation of salt suggests the circulation not as a mere territorial practice or commercial arteries but as infrastructures of political control, contestation which shapes sovereignty, dependency and spatial reordering of power along the Himalayan frontier. British colonial regulation through monopoly, taxation, and surveillance transformed long-standing Himalayan systems of mobility and exchange, embedding them within imperial circuits of extraction.

Keywords—Trade politics; Himalayan frontiers; colonial governance; commodity networks; exchange and extraction; salt economy; mobility and circulation; Named Entity Recognition; Relation extraction model

I. INTRODUCTION

The Himalayan frontier, especially Ladakh and Tibet, has long been a zone of political contestation involving the Namgyal Dynasty, the Dogras, the British administration, and the Ganden Phodrang government, along with the Chinese state [1], [2]. Yet, goods such as salt, pashm, tea, and wool have moved throughout the region with relative ease and persistence [3], [4]. These goods have acted as strategic connectors between local economies and imperial governance [5], enabling both economic sustenance and political negotiation. The Chinese annexation of Tibet and the British tax regulations have altered the nature of exchange and governance in the Western Himalayas from informal social relations, *netsang*, and barter circuits to state hierarchies over time citeMarshallBritainTibet.

The enduring salt trade circuits and the mineral's own virtue to sustain mountain economies and the shaping of rituals and practices could be studied to reveal the material foundations of governance and sovereignty in the region[6]. The paper examines how salt routes in the Western Himalayas functioned as the structural backbone of multi-commodity trade networks [7], revealing the political economy and power dynamics shaping the Himalayan salt trade in the 20th century.

The three principal salt sources of the Himalayan frontier,

i.e., the Changthang lakes harvested by Rupshus and *Changpas*; Amdo Tsaidom exploited by the Golaks and Mongols; and the Tsalkho Salt pans managed by *Khompas* and Tibetans [8], [9]. These zones have been prone to periodic political and territorial disputes, either by local groups of Kharnak (backed by Hinis Gompas) and Rupshus in Changthang or the struggles against dominance of the Ganden Phodrang government in the markets [10]. These instances prove that salt has never been a neutral commodity and that it must be studied through the lens of political economy and infrastructural power.

This paper therefore argues that salt is not a mere tradable object, it is an infrastructure of governance [11]. It aims to solidify the salt route as a political, multi-commodity trade network through computational analysis. Language processing methods such as Named Entity Recognition and relational extraction models can be employed on different historical archives, colonial gazetteers and administrative reports to construct a graph and establish salt as a political and multicommodity network.

II. HISTORICAL POLITICAL CONTEXT OF HIMALAYAN SALT

Ladakh, located along the Himalayan Karakoram arc, and adjoining the high plateaus of Western Tibet, majorly forms

the Himalayan salt frontier [12]. These high-altitude cross-roads connecting the trade routes between Tibet, Kashmir, and North India have functioned as systems of movement, kin relationship, reciprocity, and negotiation [4]. The salt lakes of Tso Kar in the Changthang plateau have sustained both the *Drokpa* herders and the Rupshu nomads who traded salt and pashm in exchange for grains, barley, apricots, and spices from lower Ladakh and Zaskar [8]. For these communities, the lakes are divine entities representing the heart of the living Buddha, their deity [8].

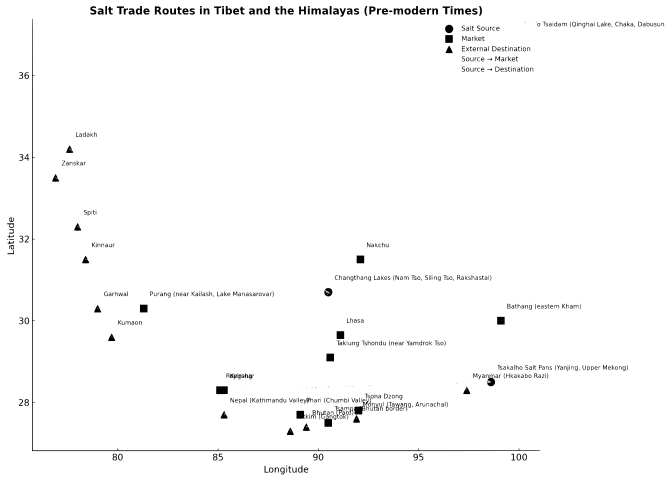


Fig. 1. A map of pre-modern salt trade routes, showing the key salt sources (circles), markets (squares), and external destinations (triangles) across the Himalayas [cite: 4, 5, 6, 7]. This illustrates the geographical scope of the networks connecting Ladakh, the Changthang Lakes, and Rupshu (Rangehar).

Before colonial interventions, salt moved through reciprocal circuits, and successive bartering was a part of the precise calendar as a function of climatic conditions and perfectly regulated trade routes [13]. The pastoral nomads, Jada, *Khompas*, and *Changpas*, along with their salt and sheep flocks, travelled up to Baltistan via Chevri-Sakti Valley and Tingmosgang in order to secure grains and apricots [4], [7]. The annual Dazem fair in Himachal provided markets for tea and sugar. In contrast, the Rupshus also silently traded with *Sbas Lung Me* (the hidden civilization) for food grains [14].

The concept of *netsang*, or fictive kinship, where each family had trade partners in another specific family in other groups that served as hosts as well, illustrates how barter operated within a moral economy of balance rather than the capitalist motives of profit [6]. TRecords of Rupshu–Kharnak disputes over control of the Tso Kar lakes, and the annual supply of salt to monasteries at Karzok and Thugji, reveal that such conflicts were resolved locally through dialogue and customary codes rather than rigid bureaucracy [10].

The Treaty of Tingmosgang of 1684 established salt as an instrument of political control long before the colonial interventions [15]. The treaty granted exclusive rights over the Ngarees Khorsum wool trade to Ladakhi merchants and restricted access to Rudok to the king’s royal traders. During the 19th century, salt exchange shifted completely from being a local affair to a matter of state monopoly owing to the pressure developed by the Chinese annexation of Tibet and British Indian expansion [1]. The arrangement for the sale of the Mundi Salt mines with the Raja of Mundi [16], [17] was an early phase of bureaucratic consolidation [18]. Salt thus became a formal object of administration that required authorization and licensing, effectively regulated by the colonial government [19], [20], [?].

This transformation of salt as a political economy under the British Rule was possible by converting the fluid frontier into a legible and taxable territory [1]. Though the cadastral surveys, route mapping, and commodity taxation brought the Himalayas one step closer to its integration into India and its economy, it turned circulation of salt into a tool of governance. According to Punjab Gazetteers and the Mandi salt reports, salt production was institutionalized, under licensing systems, fixed quotas, and surveillance checkpoints that mirrored the broader imperial Salt Department’s procedures across India.

By the twentieth century, the British fully converted the salt trade, which was once based on moral exchange and mutual recognition, into a contractual relation enforced by the state. The monetization of the trade transformed the gift into a contract, and the contract further evolved into a constraint [5]. The concept of reciprocity and *netsang* was undermined and was then overpowered by cash, duties, distant markets, and bureaucratic permissions [21].

III. METHODOLOGY

The study investigates whether salt routes functioned as the structural backbone of multicommodity trade in the Western Himalayas [7], [4]. It employs a mixed methods framework, analysing and integrating qualitative archival interpretation with quantitative text-analytic techniques. The corpus consists of colonial gazetteers [19], administrative reports [16], [17], revenue arrangements [22], [23], diplomatic treaties [15], and historical works as well [12], [6], [3]. They collectively provide valuable data on trade routes, commodity, circulation, taxation across the Himalayas.

Working on numerous NER models, the method is to integrate the Text Razor API, to enhance the named entity recognition and topic identification across the corpus. The

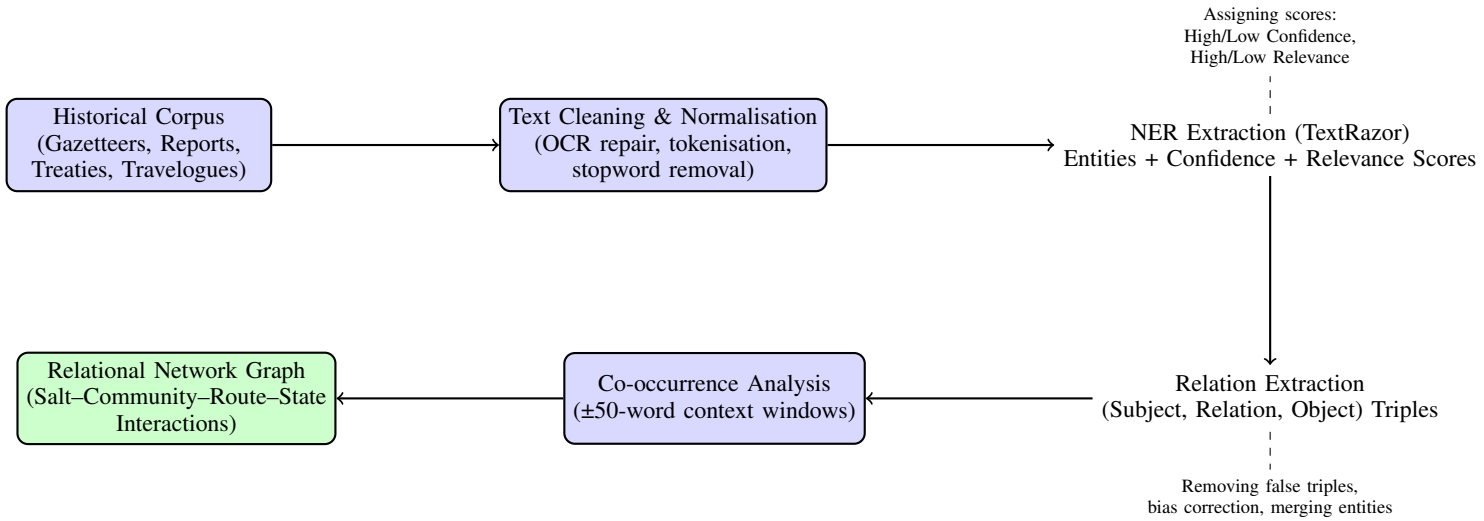


Fig. 2. **NER and Relation Extraction Pipeline used in this study.** The corpus of Himalayan historical documents was cleaned and normalised before being processed through TextRazor for Named Entity Recognition (NER). Each extracted entity received confidence and relevance scores, which allowed filtering and prioritisation. The relation extraction model was next applied to identify relational triples (e.g., “Rupshu nomads – extract salt from – Tso Kar”), followed by co-occurrence analysis based on ± 50 -word context windows. The combined outputs were then used to construct a relational network graph showing interactions between salt extraction sites, pastoral communities, trade routes, and administrative institutions. Colours used are colour-blind-friendly and the figure is fully vector-based for clarity.

entities have been categorised along with their relevance and confidence scores, further capturing the high-level topics of each document. This two-dimensional scoring helps distinguish between core trade-related concepts (high relevance and confidence) and peripheral mentions (low relevance).

Both confidence and relevance scores aim to evaluate the accuracy and importance of any extracted entity to a particular topic. The confidence score indicates how certain the model is that it has correctly identified the entity. This is especially useful in extracting rare or domain-specific entities such as Chang-Pa (people) or Zanskar (geolocation).

In contrast, relevance scores act as a tool to assess whether salt routes served as the backbone for multicommodity exchange, helping identify significant trade-related structures and practices. For example, isolating central themes such as specific commodities, trading groups, and tribes.

In this framework, high relevance and high confidence suggest strong evidence and are ideal for core analysis. High confidence but low relevance may indicate correct but minor observations and should be retained as metadata. High relevance with low confidence demands manual verification.

The next step after topic modelling and computation of scores is topic extraction. This filtering process reveals several dominant clusters including frontier governance, trade economy, pastoral communities etc. These topics provide interpretive anchors for understanding how salt is embedded with large material and political circuits.

Moving beyond the entity list and topics, the paper aims to identify relations in the form of (Subject, Relation, Object) structure within text. Examples of such relational triples include:

“Rupshu nomads — extract salt from — Tso Kar”

“Mandi State — imposes tax on — salt caravans”

“Ladakh royal trader — permitted entry to — Rudok (per Treaty of Tingmosgang)”

These relations are grounded in a word co-occurrence window approach, which uses anchor entities such as salt, wool, Rupshu, or Changthang to identify the strength and frequency of surrounding terms and their conceptual proximity. This method helps identify thematic clustering around each term. The final step requires the integration of the NER, relational triplets and co-occurrences patterns into a relational network graph. The nodes represent ecological sites, pastoral communities, state institutions, and commodities, while edges represent extraction links, taxation relations, barter dependencies, and treaty obligations.

IV. CASE STUDY: RUPSHU & TSO KAR REGION

The Tso Kar Basin and the Rupshu plateau, situated east of the lofty Ladakh along the edge of the Changthang, constitute a significant economic zone in the Himalayas [8], [24]. These high altitude saline lakes accumulate abundant salt deposits during the autumn due to evaporation [12]. These production salt beds annually attract caravans from Rupshu, Zanskar, *Drokpas* and even the Shamma Traders [4]. The Changthang lakes such as Nam Tso, Tso Kar and Seiling Tso have continued to supply salt in the Ladakh region after the Chinese annexation of Tibet, thereby sustaining long-standing circuits of exchange [1].

The Rupshu are pastoral nomads who subsist on a mixed economy of salt extraction, livestock herding, and the trade of pashm and wool [8], [25]. Their social system includes practices such as fraternal polyandry which restricts the individual households and the division of property. This also keeps

a check on the already-limited environmental and grazing sources [24]. Over time, the Rupshu pastoralism has shifted from nomadic and low intensity exploitation to sedentarization on a restricted area. This transition is contextualized by numerous 20th-century events, as shown in Figure 3.

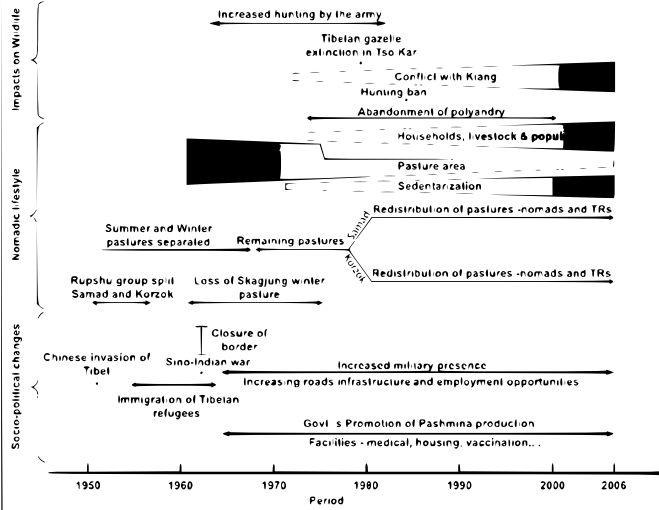


Figure 2 Chronology of events in the Changthang region from the 1950s onwards and their consequences

Fig. 3. Chronology of major socio-political and ecological events in the Changthang region from 1950. This timeline illustrates the pressures on the Rupshu nomadic lifestyle [cite: 37], including the 1959 Chinese invasion [cite: 58], border closures, and the subsequent move towards sedentarization.

The annual pastoral cycle of the *Changpas* and *Rupshus* is highly regulated, with autumn being the primary salt extraction season at Tso Kar; summer supporting mobility; spring marking birth and migration, and severe cold winters with no work [8]. Salt harvesting has been an exclusive male job. Four groups of men, *chu lag*, are led by *chu dpon* [8]. These men belong from the same *pha shun* or patri-fraternal group, reflecting the embeddedness of labour practices within kinship structures. A typical Rupshu family consists of a father, who offers prayers to deity and distributes the meat; mother who cares and cooks; a son (*bopu*) as a trainee along with Zapon the animal keeper.

Salt extraction at Tso Kar is supervised by the kotwal who is assistant of goba (village head appointed by the king). The kotwal divides the lake into four different parts and assigns

each part to the group of men based on a dice roll [6]. Each participant is required to fill a minimum of twenty five saddle bags to ensure participation and reduced grumbling. Once collected, the salt is left to dry overnight before being packed for transport.

The gendered character of salt work also shapes the linguistic and social dimensions of the trade. A distinct “salt vocabulary” circulates primarily among men, and mobility for exchange frequently depends on *ulag* (forms of obligatory or reciprocal labour) and on *netsang*, a fictive-kinship mechanism that provides hosting, protection, and trade partnerships in distant settlements [4].

Salt has functioned as a crucial element in the Rupshu life. It serves as local currency enabling the trade and acquisition of other goods such as wool, cashmere, barley and other necessities from agricultural valleys and monastic centres [13]. The Tso Kar region, with its abundant grazing fields, remains central to Rupshu pastoral subsistence. Numerous instances of control dispute between the Kharnak and the Rupshus have occurred over the past [10]. The historical volatility of salt politics in the region is further evident in wider conflicts; one widely cited instance describes the Gorkhas declaring war on Tibet over issues related to the quality of imported salt [14].

V. ANTICIPATED RESULTS

Although the computation analysis still remains under process, several anticipated results would emerge from the methods applied. Firstly, salt would appear as the central node in relation to the network graph [7]. The higher centrality will be coherent with the co-occurring commodities such as tea, pashm, grain, barley, or wool [4], [26]. This would support the central hypothesis that salt routes are the backbone for multicommodity trade.

The relation triplets are further expected to form a dense connectivity between mining sites of Tso Kar and Tsalkho Salt pans; pastoral communities of *Khompas*, *Changpas*, *Drogpas*, etc.; and several state institutions such as the British Salt Department or the Ladakhi Royal merchants [8], [9], [17], [19].

Another anticipated result can be the visibility of bottleneck zones such as the Rupshu-Tso Kar-Leh corridor, representing the intersection of extraction rights, license, seasonal mobility, and tax obligations [16], [17]. Lastly, the topic modelling would also note a gradual shift in the vocabulary. The transition would be from informal precolonial terms such as *netsang*, kinship, and reciprocity to formal colonial terms like duty, permit, monopoly, and license [21], [5].

VI. CONCLUSION

The study examined how the Western Himalayas' economies and social environments were influenced by salt as a material resource and political tool during the 20th century [11]. The study has argued that salt routes accomplished much more than just facilitating the movement of a single commodity by combining historical archives, colonial gazetteers, treaties, and computational text analysis. They served as the

backbones of infrastructure that facilitated the negotiation of state power, pastoral mobility, and regional exchange [4]. This paper has shown that salt extraction in the Rupshu and Tso Kar Basin was intertwined with territorial claims, licensing structures, and administrative interventions [17], [1], while also being ingrained in ecological rhythms, kinship, labor hierarchies, and traditional systems of reciprocity [8], [6].

Taken together, the historical trajectory from precolonial reciprocations in netsang to the contractual and bureaucratic regimes of the colonial state assigns meaning to the ways in which the Himalayan frontier gradually assumed the form of a legible and governable territory [2]. The British regulation of salt production, taxation, and mobility significantly altered the fluid frontier relationships and replaced them with a formalized regime of regulation that has remade local economies and reconstituted sovereignty at the margins of empire [18], [20]. This transformation was neither uniform nor uncontested; it interacted with pastoral strategies of adaptation, local disputes over access to lakes, and broader regional politics across Tibet, Ladakh, and colonial India [10].

A new analytical lens for interpreting the political economy of Himalayan salt is further made possible by the different computational techniques used in this study, including relational extraction, topic modeling, and NER. Expected results, like the importance of salt in network graphs or the appearance of bottleneck corridors like Rupshu–Tso Kar–Leh, give quantitative support to claims that are frequently made by historical and ethnographic scholars [27]. They highlight how Himalayan trade networks are layered, with state institutions, pastoral communities, and ecological sites forming interdependent clusters of authority and circulation. Together, these studies demonstrate that salt routes are political infrastructures that describe how frontier societies balance power, autonomy, and dependence rather than merely serving as routes for trade or transportation. The paper combines archival depth with computational modelling to advance an integrated understanding of how material substances like salt shape the territorial, social, and economic life of the Himalayas [13], [9].

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